UNIVERSITY OF BUEA

FACULTY OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF COMPUTER ENGINEERING

CEF440: INTERNET PROGRAMMING AND MOBILE PROGRAMMING

PROJECT TITLE: DESIGN AND IMPLEMENTATION OF A MOBILE APP FOR COLLECTION OF USER EXPERIENCE DATA FROM MOBILE NETWORK SUBSCRIBERS.

TASK 3: REQUIREMENT ANALYSIS

GROUP 2

COURSE INSTRUCTOR: Dr. Valery Nkemeni

S/N	NAME	MATRICULE
1.	EFUETNGONG GENESIS TAZISONG	FE23A041
2.	MATUTE MOLUA FRITZ	FE23A087
3.	NWANTOLY STEPHILL FAVOUR BENYELLA	FE23A127
4.	NYANYOH DIVINE-FORTUNE BANFILA	FE23A128
5.	TENKEN MANPLA ULRICH	FE23A170

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I. INTRODUCTION

Requirement analysis is a critical phase in the software development life cycle (SDLC) that focuses on examining, refining, validating, and organizing the requirements collected during the requirement gathering phase. The purpose of this phase is to ensure that the proposed system satisfies stakeholder expectations while remaining technically feasible, reliable, secure, and user-friendly.

In this project, requirement analysis was carried out for the proposed mobile application designed to collect Quality of Experience (QoE) data from mobile network subscribers. The application is intended to gather both subjective user feedback and objective network performance metrics in real time. These metrics include signal strength, latency, network type, call quality, internet speed, and user satisfaction ratings.

The analysis phase was necessary to transform raw information gathered from surveys, interviews, brainstorming sessions, and reverse engineering into structured system requirements that can guide the design and implementation of the application.

The requirement analysis process focused on:

- Reviewing and evaluating gathered requirements
- Identifying ambiguities and inconsistencies
- Determining technical feasibility
- Prioritizing system requirements
- Classifying requirements into functional and non-functional categories
- Developing the Software Requirement Specification (SRS)
- Validating the requirements with stakeholders

This phase serves as the foundation for subsequent phases such as system design, implementation, testing, and deployment.

II. REVIEW AND ANALYSIS OF GATHERED REQUIREMENTS

The requirements collected during Task 2 were carefully reviewed and analyzed to determine whether they were complete, understandable, achievable, and relevant to the objectives of the project.

The requirements were gathered from several sources, including:

- Online surveys using Google Forms
- Interviews with mobile subscribers
- Brainstorming sessions
- Analysis of existing network monitoring applications

The collected requirements were analyzed based on the following criteria:

i. Completeness of Requirements

Completeness analysis was performed to ensure that all necessary aspects of the proposed system were captured during the requirement gathering process. A requirement is considered complete when it fully describes the expected functionality or behavior of the system without leaving significant gaps.

The analysis confirmed that the gathered requirements sufficiently covered the following areas:

Functional Requirements

These requirements describe the operations the system must perform.

The system should:

- ✓ Collect user feedback regarding mobile network performance
- ✓ Automatically gather network performance metrics
- ✓ Store collected data in a centralized database
- ✓ Generate reports and visual analytics
- ✓ Operate continuously in the background
- ✓ Allow real-time synchronization of collected data

User Requirements

These requirements focus on user expectations and usability.

Users expect the system to:

- ✓ Have a simple and intuitive interface
- ✓ Consume minimal battery power
- ✓ Protect personal information and privacy
- ✓ Operate smoothly without interrupting device usage

- ✓ Provide meaningful insights and feedback

Technical Requirements

These requirements focus on technologies and infrastructure needed for implementation.

The application requires:

- ✓ Android platform compatibility
- ✓ Internet connectivity
- ✓ GPS/location services
- ✓ Cloud or local database integration
- ✓ Background service capabilities
- ✓ Secure data transmission mechanisms

The completeness review showed that the system requirements adequately addressed the major operational, user, and technical aspects needed for the successful implementation of the project.

ii. Clarity of Requirements

Requirement clarity is essential to avoid misunderstandings between developers, stakeholders, and users. Ambiguous or poorly defined requirements can lead to implementation errors, system failures, or unmet stakeholder expectations.

During the analysis process, all requirements were examined and refined to improve precision and understanding.

For example:

Initial Requirement

"The app should monitor the network."

This statement was too broad and lacked sufficient detail.

Refined Requirement

The application shall automatically monitor and record network parameters such as signal strength, latency, jitter, packet loss, network type, and timestamp information in real time.

The refined version clearly defines:

- What should be monitored
- Which parameters should be collected
- How the monitoring should occur
- When the monitoring should take place

Another example involved user feedback collection.

Initial Requirement

"Users should provide feedback."

Refined Requirement

The application shall periodically prompt users to rate network quality, call quality, internet speed, and overall satisfaction using a structured rating system.

This refinement eliminated ambiguity and improved implementation guidance.

As a result, all requirements were rewritten in clear, measurable, and understandable language to facilitate accurate system development.

iii. Technical Feasibility Analysis

Technical feasibility analysis was conducted to determine whether the proposed system could realistically be implemented using available technologies, tools, resources, and development methodologies.

The feasibility study examined several important aspects.

Mobile Platform Feasibility

The Android platform was selected because:

- Android devices are widely used among mobile subscribers
- Android provides APIs for network monitoring
- Android supports background services and sensor integration
- Development tools for Android are readily available

Therefore, the platform was considered highly feasible for implementation.

Data Collection Feasibility

The proposed system requires automatic collection of network and device information such as:

- Signal strength
- Network type
- Internet speed
- GPS location
- Timestamps

These features are achievable through:

- Android Telephony APIs
- Network monitoring libraries
- GPS services
- Internet connectivity services

Hence, automated data collection was found to be technically feasible.

Database Feasibility

The application requires reliable storage of large volumes of collected data.

Possible solutions include:

- Firebase Realtime Database
- MySQL
- SQLite
- Cloud-based storage systems

These technologies provide:

- Secure storage
- Real-time synchronization
- Scalability
- Data accessibility

Database implementation was therefore considered feasible.

Performance Feasibility

One major concern involved background operation and battery consumption.

Continuous monitoring can increase CPU usage, Drain battery power, Consume internet data

To address this issue:

- Data collection intervals can be optimized
- Background tasks can be scheduled efficiently
- Lightweight monitoring techniques can be implemented

The analysis concluded that acceptable performance can be achieved with proper optimization techniques.

Security Feasibility

The system will collect potentially sensitive user information such as:

- Location data
- Device information
- Network usage patterns

To ensure security:

- Data encryption techniques will be implemented
- User permissions will be requested transparently
- Secure communication protocols will be used

Therefore, secure implementation of the system is feasible.

iv. Dependency Relationships

Dependency analysis was performed to identify relationships between system components and requirements. Certain functionalities depend on the availability or successful operation of others.

Understanding these relationships helps developers plan implementation properly.

Identified Dependencies

Requirement	Dependency
Real-time data synchronization	Internet connectivity
Location-based monitoring	GPS permission
Network performance tracking	Android device APIs
Cloud data storage	Database server availability
User feedback submission	User interaction
Report generation	Stored and processed data

For example:

- The application cannot perform location analysis unless GPS access is granted.
- Real-time synchronization cannot function without internet access.
- Report generation depends on previously stored data.

Recognizing these dependencies helps avoid system integration problems during development.

III. IDENTIFICATION OF INCONSISTENCIES, AMBIGUITIES, AND MISSING INFORMATION

During this analysis phase, several inconsistencies, ambiguities, and missing requirements were identified and addressed.

i. Inconsistencies

Inconsistencies were mainly observed in survey responses due to differences in user input styles.

Examples included:

- “mtn”, “MTN”, and “Mtn”
- “iphone” and “iPhone”
- “Orange Cameroon” and “Orange”

These inconsistencies could negatively affect statistical analysis and data visualization.

To solve this problem:

- Standardized naming conventions were applied
- Duplicate entries were removed
- Formatting rules were enforced

This ensured consistency throughout the dataset.

ii. Ambiguities

Some responses lacked sufficient detail for accurate interpretation.

For example,
Several users rated network quality as “Poor” without specifying whether the issue was related to:

- Internet speed
- Signal strength
- Call quality
- Network stability

To resolve these ambiguities:

- Additional survey questions were introduced
- More detailed rating categories were added
- Structured response options were implemented

This improved data accuracy and interpretability.

iii. Missing Information

The analysis identified several important features that were not initially captured during requirement gathering.

Missing requirements included:

- User authentication mechanisms
- Data encryption and security policies
- Offline data storage support
- Notification management
- Error handling mechanisms
- Data backup and recovery

These requirements were added to strengthen the system design.

IV. REQUIREMENT PRIORITIZATION

After analysis, the requirements were prioritized based on:

- Importance to system functionality
- User impact
- Technical feasibility
- Development complexity

This prioritization helps guide implementation planning.

I. High Priority Requirements

These are essential features without which the system cannot function properly.

Requirement	**Importance**
User feedback collection	Core functionality
Network monitoring	Essential for QoE analysis
Data storage	Required for analysis
Background operation	Main project objective
Privacy protection	Critical for user trust

II. Medium Priority Requirements

These features improve usability and system effectiveness.

Requirement	**Importance**
GPS integration	Improves network analysis
Notifications	Enhances interaction
Data visualization	Supports decision-making
Cloud synchronization	Enables remote access

III. Low Priority Requirements

These are optional enhancements.

Requirement	**Importance**
Dark mode	Cosmetic feature
Theme customization	User preference
Personalized dashboards	Additional convenience

IV. CLASSIFICATION OF REQUIREMENTS

The analyzed requirements were classified into Functional requirements and Non-functional requirements

i. Functional Requirements

Functional requirements describe what the system must do.

The application shall:

- Allow users to submit feedback
- Monitor network quality automatically
- Collect signal strength and latency data
- Store user and network data
- Generate reports and charts
- Synchronize collected data with a server
- Record timestamps and locations
- Operate continuously in the background

ii. Non-Functional Requirements

Non-functional requirements define the quality attributes of the system.

Performance Requirements

- The application should respond quickly
- Data collection should occur efficiently
- Battery consumption should remain low

Security Requirements

- User information must be protected
- Data transmission should be encrypted
- Access permissions must be controlled

Reliability Requirements

- The application should run continuously with minimal failures
- Data loss should be minimized

Usability Requirements

- The interface should be simple and intuitive
- Users should easily understand feedback prompts

Compatibility Requirements

- The application should support Android devices
- It should function across multiple Android versions

V. SOFTWARE REQUIREMENT SPECIFICATION (SRS)

The Software Requirement Specification provides a detailed description of the proposed system and serves as a guide for developers during implementation.

i. System Overview

The proposed system is a mobile application designed to collect and analyze user experience data related to mobile network performance. The system combines automated monitoring with user feedback to provide accurate Quality of Experience analysis.

ii. System Inputs

The system will collect:

- User satisfaction ratings
- Signal strength
- Network type
- Internet speed
- Call quality information
- GPS location
- Device information
- Timestamp data

iii. System Outputs

The system will generate:

- Statistical reports
- Charts and graphs
- QoE analysis summaries
- Network performance reports
- User satisfaction analytics

iv. Major System Modules

The system will consist of:

- User Interface Module
- Network Monitoring Module
- Data Collection Module
- Database Management Module
- Reporting and Analytics Module

VI. VALIDATION OF REQUIREMENTS WITH STAKEHOLDERS

Requirement validation was performed to confirm that the analyzed requirements accurately reflect stakeholder expectations and project objectives.

Validation activities included:

- Reviewing survey findings
- Discussing proposed features with users
- Confirming technical feasibility with developers
- Verifying privacy and usability concerns

Stakeholders agreed that the proposed system would:

- ✓ Improve network quality monitoring
- ✓ Enhance customer satisfaction analysis
- ✓ Support informed decision-making by network operators

The validation process ensured that the final requirements are realistic, relevant, and implementable.

VII. CONCLUSION

The requirement analysis phase successfully transformed raw data and stakeholder feedback into structured, validated, and prioritized system requirements. Through detailed analysis, inconsistencies and ambiguities were resolved, missing requirements were identified, and the technical feasibility of the system was confirmed.

The analysis established a strong foundation for:

- System architecture design
- Database design
- Mobile application development
- QoE analytics implementation

By completing this phase, the project now possesses a clear roadmap for developing a reliable, secure, efficient, and user-centered mobile application capable of collecting and analyzing user experience data from mobile network subscribers in real time.